Barcelona Port and Zona Franca

The Problem -
Traffic is the main source of toxic particles, accounting for 20% of NO2 levels, with construction dust and shipping emissions also contributing.

The Culprit -
Chemical pollutants, including industrial perfluorinated compounds, microplastics, and pharmaceutical residues, are detected in, sewage and groundwater, impacting aquatic organisms and potentially water safety.

The Solution -
The Mediterranean Action Plan (MAP), established by UNEP, is a regional framework designed to protect the Mediterranean Sea from pollution while supporting sustainable coastal development. Its legal backbone is the Barcelona Convention, which commits Mediterranean countries to prevent, reduce, and control marine pollution from land-based sources, maritime traffic, and industrial activity. Through binding protocols, shared environmental targets, and coordinated monitoring, MAP links scientific research with policy enforcement—though its effectiveness is often challenged by intense port activity, industrial zones, and uneven implementation at local scales, such as major hubs like the Port of Barcelona.

Action Plan -
MAP supports Mediterranean countries in implementing the Barcelona Convention by helping translate international agreements into national policies and on-the-ground actions. Most member states have adopted National Action Plans and coastal management strategies that commit to reducing marine pollution, protecting biodiversity, and promoting sustainable coastal development.

Interesting facts about the Mediterranean -
The Mediterranean is one of the world's 25 hot spots for biodiversity. Its highly diverse marine ecosystem hosts around 4 to 18% of the world's marine biodiversity.
The Mediterranean Sea is the largest semi-enclosed sea in the world with 46,000 km of coastline.
70% of the Mediterranean population lives in urban areas; one in three lives in Mediterranean coastal areas.
The Mediterranean Sea is host to the world's busiest shipping lanes and is the second busiest cruising region in the world after the Caribbean with 15.8 per cent of global cruise fleet deployment in 2017 (MedCruise Association, 2018).
The Mediterranean region features one of the highest amounts of municipal solid waste generated annually per person: 208 – 760 kg/year.
Marine litter in the Mediterranean is mostly composed of plastics and concentrations are high: more than 100,000 microplastic items/km2, and up to 64 million particles/ km2 of floating litter. 0.5 billion items are lying on the Mediterranean seafloor with densities sometimes exceeding 100,000 items/km2.

The Mediterranean region has warmed approx. 1.5°C since pre-industrial times, 20% faster than the global average, according to MedECC.





